

Online Supplement to

Strategic Analysis of Travel Restriction Policies during a Pandemic

Appendix S1: Proof of Propositions A1, A2 and Proposition A3 in the Appendix

S1.1. Proof of Proposition A1 and A2

Proof. Since $\hat{k}$ is uniquely determined by the intersection of the series $\{s_i\}_{i=\bar{k}+1}^n$ and the series $\{\Lambda_i\}_{i=\bar{k}+1}^n$, it's sufficient to observe the movement of the curve $\{(i, \Lambda_i)\}_{i=\bar{k}+1}^n$ to determine the change in $\hat{k}$. Λ_i is nothing but a weighted average of s_0 and s_j 's ($j = i, i+1, \dots, n$), with the corresponding weights $1 - \sum_{j=1}^{\bar{k}} N_{0j}$ and N_{j0} 's ($j = i, i+1, \dots, n$) normalized by $1 - \sum_{j=1}^{\bar{k}} N_{0j} + \sum_{j=i}^n N_{j0}$. Thus, the increase in N_{0i} ($i = 1, 2, \dots, k$) will shift the curve downward, causing $\hat{k}$ to increase; but nothing changes when N_{0i} ($i = \bar{k}+1, \dots, n$) varies. On the other hand, the increase in a typical N_{i0} where $i = \hat{k}+1, \dots, n$, would augment the series $\{\Lambda_j\}_{j=\bar{k}+1}^i$ to decrease and thus cause $\hat{k}$ to increase. $\hat{k}$ does not move in correspondence to any change in N_{i0} ($i = 1, \dots, \hat{k}$).

The direct impact of the incorporation of a city with $s \geq \Lambda_{\hat{k}+1}$ is that $\hat{k}$ must increase by at least one unit. When $s > s_0$, in calculating Λ_i for a particular i , the weight of s_0 decreases and the weights for $s_i, \dots, s_n$ increase. As a consequence, Λ_i decreases and the curve of $\{\Lambda_i\}_{i=\bar{k}+1}^n$ will move downwards, causing the intersection point to move to the right and thus $\hat{k}$ to remain unchanged or increase. With the index shift, $\Delta\hat{k}$ may be larger than 1. $\square$

S1.2. Proof of Proposition A3

Proof. • Case 1: $s_1 = s_0$. When $s_1 = s_2 = s_0$, one can solve that $(X_j^*, Y_j^*) = (0, 0)$, since the first-order derivatives are all negative given that $A > 0$. Thus, $a_j^* = z$ and $b_j^* = z$.

When $s_1 = s_0 \neq s_2$, we can solve $a_1^* = z$, namely $y_1^* = 0$. If $X_1^* = 0$, it suggests that the population sizes in the center city 0 and city 1 after the travel do not change. One can check that $\frac{m-m^2}{m+1} < 1/4$ always holds. Thus the optimum (X_2^*, Y_2^*) is exactly characterized as in Theorem 9. On the other hand, if $X_1^* = 1$, the population size in city 0 becomes $1 - z$. Depending on whether $1 - z > m$ given that $z \leq \min\{\frac{1}{4}, \frac{m}{2}\}$, we can characterize the optimum (X_2^*, Y_2^*) as follows,

— when $\frac{1}{m+z} - \frac{1}{2m} - \frac{1}{2(1-z)} > 0$ and $z \leq \min\{\frac{1}{4}, \frac{m}{2}\}$, the social optimum is:

$$(X_2^*, Y_2^*) = \begin{cases} (1, 1), & \text{if } A \leq \beta(s_2 - s_0)^2 \frac{m}{m+z}; \\ (1, 0), & \text{if } \beta(s_2 - s_0)^2 \frac{m}{m+z} < A \leq \beta(s_2 - s_0)^2 (\frac{m+2z}{m+z} - \frac{z}{m} - \frac{z}{1-z}); \\ (0, 0), & \text{if } A > \beta(s_2 - s_0)^2 (\frac{m+2z}{m+z} - \frac{z}{m} - \frac{z}{1-z}). \end{cases}$$

— when $\frac{1}{m+z} - \frac{1}{2m} - \frac{1}{2(1-z)} < 0$ and $z \leq \min\{\frac{1}{4}, \frac{m}{2}\}$, the social optimum is:

$$(X_2^*, Y_2^*) = \begin{cases} (1, 1), & \text{if } A \leq \beta(s_2 - s_0)^2 (1 - \frac{z}{2m} - \frac{z}{2(1-z)}); \\ (0, 0), & \text{if } A > \beta(s_2 - s_0)^2 (1 - \frac{z}{2m} - \frac{z}{2(1-z)}). \end{cases}$$

Given the characterization of (X_2^*, Y_2^*) , we further need to check what is X_1^* under a specific pair of (X_2^*, Y_2^*) . Actually, we only need to check the five thresholds $T_0 := \frac{m}{m+z}$, $T_3 = 1 - z - \frac{z^2}{m(m+z)}$, $T_4 := 1 - \frac{m+1}{2m}z$, $T_5 := \frac{m+2z}{m+z} - \frac{z}{m} - \frac{z}{1-z}$, and $T_6 := 1 - \frac{z}{2m} - \frac{z}{2(1-z)}$. The relationship between the five thresholds is: $T_6 - T_4 = -\frac{z^2}{2(1-z)} < 0$, $T_5 - T_3 = -\frac{z^2}{1-z} < 0$, and $T_0 + T_5 = 2T_6$. Thus, we conclude that, when $T_0 < T_5 < T_3$, the range of A is divided into $(-\infty, T_0] \cup (T_0, T_5] \cup (T_5, T_3] \cup (T_3, \infty)$; when $T_5 < T_0 < T_3$, the range of A is divided into $(-\infty, T_6] \cup (T_6, T_0] \cup (T_0, T_3] \cup (T_3, \infty)$; when $T_5 < T_3 < T_0$, the range of A is divided into $(-\infty, T_6] \cup (T_6, T_4] \cup (T_4, \infty)$. Combining the corresponding optimal strategies (X_2^*, Y_2^*) , we summarize the candidate optimal strategies as follows:

— when $\frac{1}{m+z} - \frac{1}{2m} - \frac{1}{2(1-z)} > 0$, the social optimum

$$(X_1^*, X_2^*, Y_2^*) \in \begin{cases} \{(\mathbf{0}, \mathbf{1}, \mathbf{1}), (1, 1, 1)\}, & \text{if } A \leq \beta(s_2 - s_0)^2 \frac{m}{m+z}; \\ \{(\mathbf{0}, \mathbf{1}, \mathbf{0}), (1, 1, 0)\}, & \text{if } \beta(s_2 - s_0)^2 \frac{m}{m+z} < A \leq \beta(s_2 - s_0)^2 (\frac{m+2z}{m+z} - \frac{z}{m} - \frac{z}{1-z}); \\ \{(\mathbf{0}, \mathbf{1}, \mathbf{0}), (1, 0, 0)\}, & \text{if } \beta(s_2 - s_0)^2 (\frac{m+2z}{m+z} - \frac{z}{m} - \frac{z}{1-z}) < A \leq \beta(s_2 - s_0)^2 (1 - z - \frac{z^2}{m(m+z)}); \\ \{(\mathbf{0}, \mathbf{0}, \mathbf{0}), (1, 0, 0)\}, & \text{if } A > \beta(s_2 - s_0)^2 (1 - z - \frac{z^2}{m(m+z)}). \end{cases}$$

— when $\frac{1}{m+z} - \frac{1}{2m} - \frac{1}{2(1-z)} < 0$ and $z < \frac{m-m^2}{m+1}$, the social optimum

$$(X_1^*, X_2^*, Y_2^*) \in \begin{cases} \{(\mathbf{0}, \mathbf{1}, \mathbf{1}), (1, 1, 1)\}, & \text{if } A \leq \beta(s_2 - s_0)^2 (1 - \frac{z}{2m} - \frac{z}{2(1-z)}); \\ \{(\mathbf{0}, \mathbf{1}, \mathbf{1}), (1, 0, 0)\}, & \text{if } \beta(s_2 - s_0)^2 (1 - \frac{z}{2m} - \frac{z}{2(1-z)}) < A \leq \beta(s_2 - s_0)^2 \frac{m}{m+z}; \\ \{(\mathbf{0}, \mathbf{1}, \mathbf{0}), (1, 0, 0)\}, & \text{if } \beta(s_2 - s_0)^2 \frac{m}{m+z} < A \leq \beta(s_2 - s_0)^2 (1 - z - \frac{z^2}{m(m+z)}); \\ \{(\mathbf{0}, \mathbf{0}, \mathbf{0}), (1, 0, 0)\}, & \text{if } A > \beta(s_2 - s_0)^2 (1 - z - \frac{z^2}{m(m+z)}). \end{cases}$$

— when $\frac{1}{m+z} - \frac{1}{2m} - \frac{1}{2(1-z)} < 0$ and $z > \frac{m-m^2}{m+1}$, the social optimum

$$(X_1^*, X_2^*, Y_2^*) \in \begin{cases} \{(\mathbf{0}, \mathbf{1}, \mathbf{1}), (1, 1, 1)\}, & \text{if } A \leq \beta(s_2 - s_0)^2 (1 - \frac{z}{2m} - \frac{z}{2(1-z)}); \\ \{(\mathbf{0}, \mathbf{1}, \mathbf{1}), (1, 0, 0)\}, & \text{if } \beta(s_2 - s_0)^2 (1 - \frac{z}{2m} - \frac{z}{2(1-z)}) < A \leq \beta(s_2 - s_0)^2 (1 - \frac{m+1}{2m}z); \\ \{(\mathbf{0}, \mathbf{0}, \mathbf{0}), (1, 0, 0)\}, & \text{if } A > \beta(s_2 - s_0)^2 (1 - \frac{m+1}{2m}z). \end{cases}$$

The optimal strategies are indicated in bold letters. We outline the proof in the following. When $X_2^* = Y_2^* = 1$, the minimum of the objective is obtained when $b_1 = z$ and $X_1^* = 0$.

Let's compare the objective value when $(X_1, X_2, Y_2) = (0, 1, 1)$ with that of $(1, 0, 0)$. One can check that $(X_1^*, X_2^*, Y_2^*) = (0, 1, 1)$ if and only if $A < \beta(s_2 - s_0)^2 (2 - \frac{z}{m} - \frac{z}{1-z})$, which fits into the range of A . Thus, $(X_1, X_2, Y_2) = (0, 1, 1)$ when compared with $(1, 0, 0)$.

The comparison between the objective values when $(X_1^*, X_2^*, Y_2^*) = (0, 1, 0)$ with that of $(1, 1, 0)$ shows that, the objective value for $(X_1, X_2, Y_2) = (0, 1, 0)$ is always lower, suggesting that $(X_1^*, X_2^*, Y_2^*) = (0, 1, 0)$ when compared with $(1, 1, 0)$.

Next, we compare the objective value when $(X_1, X_2, Y_2) = (0, 1, 0)$ with that of $(1, 0, 0)$. $(X_1^*, X_2^*, Y_2^*) = (0, 1, 0)$ if and only if $mz + m^2 - z^2 > 0$. The inequality always holds since we have $m > z$. Thus, $(X_1^*, X_2^*, Y_2^*) = (0, 1, 0)$ when compared with $(1, 0, 0)$.

When $X_2^* = Y_2^* = (0, 0)$, the term related to b_1 in the objective is $-\beta(s_2 - s_0)^2 \frac{z^2}{1-z+b_1} - Ab_1$. One can check that $X_1^* = 0$ if and only if $A > \beta(s_2 - s_0)^2 \frac{z^2}{1-z}$. First, we compare $\frac{z^2}{1-z}$ and $1 - \frac{m+1}{2m}z$. $\frac{z^2}{1-z} < 1 - \frac{m+1}{2m}z$ is equivalent to $(1-m)z^2 - (3m+1)z + 2m > 0$, which always holds when $\frac{m-m^2}{m+1} < z < \frac{m}{2}$. That is, $A > \beta(s_2 - s_0)^2 (1 - \frac{m+1}{2m}z)$ implies $A > \beta(s_2 - s_0)^2 \frac{z^2}{1-z}$. Thus $X_1^* = 0$ if $A > \beta(s_2 - s_0)^2 (1 - \frac{m+1}{2m}z)$. Second, we compare $\frac{z^2}{1-z}$ with $1 - z - \frac{z^2}{m(m+z)}$. When $\frac{1}{m+z} - \frac{1}{2m} - \frac{1}{2(1-z)} > 0$, $1 - z - \frac{z^2}{m(m+z)} - \frac{z^2}{1-z} = \frac{m+2z}{m+z} - \frac{z}{m} - \frac{z}{1-z} > \frac{m}{m+z} > 0$, which suggests that $A > \beta(s_2 - s_0)^2 (1 - z - \frac{z^2}{m(m+z)})$ implies $A > \beta(s_2 - s_0)^2 \frac{z^2}{1-z}$. Thus $X_1^* = 0$. Otherwise when $\frac{1}{m+z} - \frac{1}{2m} - \frac{1}{2(1-z)} < 0$, $1 - z - \frac{z^2}{m(m+z)} - \frac{z^2}{1-z} = \frac{m^2+m(1-m)z-(m+1)z^2}{m(m+z)} > 0$ given that $0 < z < \frac{m-m^2}{m+1}$. Thus, we can conclude that $(X_1^*, X_2^*, Y_2^*) = (0, 0, 0)$ when compared with $(1, 0, 0)$.

- Case 2: $s_0 = s_2$. Similar to Case 1, we can solve $Y_2^* = 0$ and $X_2^* = 0$, and the optimum (X_1^*, Y_1^*) is exactly characterized in Theorem 9.

- Case 3: $s_1 = s_2 \neq s_0$. The objective is equivalent to

$$\min_{a_1, a_2, b_1, b_2} -\beta(s_1 - s_0)^2 \left[\frac{(m-b_1)^2}{m-b_1+a_1} + \frac{(m-b_2)^2}{m-b_2+a_2} + \frac{(b_1+b_2)^2}{1-(a_1+a_2)+(b_1+b_2)} \right] - A(a_1+a_2) - A(b_1+b_2).$$

For simplicity, we denote the objective function as $C(a_1, a_2, b_1, b_2)$. Notice that if $b_1 = b_2 = 0$, then $a_1^* = a_2^*$ and $a_1^* = 0$ if and only if $A < \beta(s_1 - s_0)^2 \frac{m}{m+z}$. That means, $(a_1, a_2, b_1, b_2) = (0, 0, 0, 0)$ is one candidate optimal strategy when $A < \beta(s_1 - s_0)^2 \frac{m}{m+z}$, and $(a_1, a_2, b_1, b_2) = (z, z, 0, 0)$ is one candidate optimal strategy when $A > \beta(s_1 - s_0)^2 \frac{m}{m+z}$.

If $b_1 = 0$ and $b_2 = z$, we make the following observations:

- If $a_1 = 0$, $a_2^* = 0$ if and only if $A < \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{z^2}{1+z})$.
- If $a_1 = z$, $a_2^* = 0$ if and only if $A < \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{z^2}{1+z})$.
- If $a_2 = 0$, $a_1^* = 0$ if and only if $A < \beta(s_1 - s_0)^2(\frac{m}{m+z} - \frac{z^2}{1+z})$.
- If $a_2 = z$, $a_1^* = 0$ if and only if $A < \beta(s_1 - s_0)^2(\frac{m}{m+z} - \frac{z^2}{1+z})$.

Similarly, if $b_1 = z$ and $b_2 = 0$, we can analyze the optimal strategies correspondingly.

If $b_1 = b_2 = z$, we have:

- If $a_1 = 0$, $a_2^* = 0$ if and only if $A < \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{(1+z)(1+2z)})$.
- If $a_1 = z$, $a_2^* = 0$ if and only if $A < \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{(1+z)(1+2z)})$.
- If $a_2 = 0$, $a_1^* = 0$ if and only if $A < \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{(1+z)(1+2z)})$.
- If $a_2 = z$, $a_1^* = 0$ if and only if $A < \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{(1+z)(1+2z)})$.

The analysis over the thresholds of A shows that the inequalities $1 - \frac{z}{m} - \frac{4z^2}{(1+z)(1+2z)} < 1 - \frac{z}{m} - \frac{z^2}{1+z} < 1 - \frac{z}{m} - \frac{z^2}{1+z} < \frac{m}{m+z} - \frac{z^2}{1+z} < \frac{m}{m+z} - \frac{z^2}{1+z} < \frac{m}{m+z}$ holds. Thus, we summarize the candidate socially optimal strategies as follows:

$$(a_1^*, a_2^*, b_1^*, b_2^*) \in \begin{cases} \{(0, 0, z, z), (0, 0, z, 0), (0, 0, 0, 0)\}, & \text{if } A \leq \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{1+2z}); \\ \{(z, z, z, z), (0, 0, z, 0), (0, 0, 0, 0)\}, & \text{if } \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{1+2z}) < A \leq \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{z^2}{1+z}); \\ \{(z, z, z, z), (z, 0, z, 0), (0, 0, 0, 0)\}, & \text{if } \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{z^2}{1+z}) < A \leq \beta(s_1 - s_0)^2(\frac{m}{m+z} - \frac{z^2}{1+z}); \\ \{(z, z, z, z), (z, z, z, 0), (0, 0, 0, 0)\}, & \text{if } \beta(s_1 - s_0)^2(\frac{m}{m+z} - \frac{z^2}{1+z}) < A \leq \beta(s_1 - s_0)^2\frac{m}{m+z}; \\ \{(z, z, z, z), (z, z, z, 0), (z, z, 0, 0)\}, & \text{if } A > \beta(s_1 - s_0)^2\frac{m}{m+z}. \end{cases}$$

Next, we will identify the unique socially optimal strategy when A falls within some range.

When $A \leq \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{1+2z})$, $C(0, 0, 0, 0) - C(0, 0, z, 0) = z(A - \beta(s_1 - s_0)^2\frac{1}{1+z}) < 0$. $C(0, 0, 0, 0) - C(0, 0, z, z) = 2z(A - \beta(s_1 - s_0)^2\frac{1}{1+2z})$. Thus,

— if $1 - \frac{z}{m} - \frac{4z^2}{1+2z} < \frac{1}{1+2z}$ (which is equivalent to $z > \frac{2m-1}{4m+2}$), then $A \leq \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{1+2z}) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (0, 0, 0, 0)$;

— otherwise, if $z < \frac{2m-1}{4m+2}$, then $A \leq \beta(s_1 - s_0)^2\frac{1}{1+2z} \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (0, 0, 0, 0)$, and $\beta(s_1 - s_0)^2\frac{1}{1+2z} < A \leq \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{1+2z}) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (0, 0, z, z)$.

When $\beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{1+2z}) < A \leq \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{z^2}{1+z})$, $C(0, 0, 0, 0) - C(0, 0, z, 0) = z(A - \beta(s_1 - s_0)^2\frac{1}{1+z}) < 0$. $C(0, 0, 0, 0) - C(z, z, z, z) = 4z(A - \beta(s_1 - s_0)^2(1 - z - \frac{z}{2m}))$. Thus,

— if $1 - z - \frac{z}{2m} < 1 - \frac{z}{m} - \frac{4z^2}{1+2z}$ (which is equivalent to $z < \frac{2m-1}{4m+2}$), then $\beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{1+2z}) < A \leq \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{z^2}{1+z}) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (z, z, z, z)$;

— if $1 - \frac{z}{m} - \frac{4z^2}{1+2z} < 1 - z - \frac{z}{2m} < 1 - \frac{z}{m} - \frac{z^2}{1+z}$, then $\beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{1+2z}) < A \leq \beta(s_1 - s_0)^2(1 - z - \frac{z}{2m}) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (0, 0, 0, 0)$, and $\beta(s_1 - s_0)^2(1 - z - \frac{z}{2m}) < A \leq \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{z^2}{1+z}) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (z, z, z, z)$;

— if $1 - \frac{z}{m} - \frac{z^2}{1+z} < 1 - z - \frac{z}{2m}$, then $\beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{4z^2}{1+2z}) < A \leq \beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{z^2}{1+z}) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (0, 0, 0, 0)$.

When $\beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{z^2}{1+z}) < A \leq \beta(s_1 - s_0)^2(\frac{m}{m+z} - \frac{z^2}{1+z})$, $C(0, 0, 0, 0) - C(z, 0, z, 0) = 2z(A - \beta(s_1 - s_0)^2(1 - \frac{z}{2} - \frac{z}{2m}))$, $C(0, 0, 0, 0) - C(z, z, z, z) = 4z(A - \beta(s_1 - s_0)^2(1 - z - \frac{z}{2m}))$, $C(z, 0, z, 0) - C(z, z, z, z) = 2z(A - \beta(s_1 - s_0)^2(1 - \frac{3z}{2} - \frac{z}{2m}))$. Since $1 - \frac{3z}{2} - \frac{z}{2m} < 1 - z - \frac{z}{2m} < 1 - \frac{z}{2} - \frac{z}{2m}$, we can conclude that as long as $A \leq \beta(s_1 - s_0)^2(1 - z - \frac{z}{2m})$, $(a_1^*, a_2^*, b_1^*, b_2^*) = (0, 0, 0, 0)$; otherwise, $(a_1^*, a_2^*, b_1^*, b_2^*) = (z, z, z, z)$. Next, we compare $1 - z - \frac{z}{2m}$ with the bounds of the range of A and summarize the following properties:

— if $1 - z - \frac{z}{2m} < 1 - \frac{z}{m} - \frac{z^2}{1+z}$, then $\beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{z^2}{1+z}) < A \leq \beta(s_1 - s_0)^2(\frac{m}{m+z} - \frac{z^2}{1-z}) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (z, z, z, z)$;

— if $1 - \frac{z}{m} - \frac{z^2}{1+z} < 1 - z - \frac{z}{2m} < \frac{m}{m+z} - \frac{z^2}{1-z}$, then $\beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{z^2}{1+z}) < A \leq \beta(s_1 - s_0)^2(1 - z - \frac{z}{2m}) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (0, 0, 0, 0)$, and $\beta(s_1 - s_0)^2(1 - z - \frac{z}{2m}) < A \leq \beta(s_1 - s_0)^2(\frac{m}{m+z} - \frac{z^2}{1-z}) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (z, z, z, z)$;

— if $\frac{m}{m+z} - \frac{z^2}{1-z} < 1 - z - \frac{z}{2m}$, then $\beta(s_1 - s_0)^2(1 - \frac{z}{m} - \frac{z^2}{1+z}) < A \leq \beta(s_1 - s_0)^2(\frac{m}{m+z} - \frac{z^2}{1-z}) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (0, 0, 0, 0)$.

When $\beta(s_1 - s_0)^2(\frac{m}{m+z} - \frac{z^2}{1-z}) < A \leq \beta(s_1 - s_0)^2\frac{m}{m+z}$, $C(0, 0, 0, 0) - C(z, z, z, 0) = 3z(A - \beta(s_1 - s_0)^2(1 - \frac{z}{3m} - \frac{z}{3(1-z)} - \frac{z}{3(m+z)}))$, $C(0, 0, 0, 0) - C(z, z, z, z) = 4z(A - \beta(s_1 - s_0)^2(1 - z - \frac{z}{2m}))$, $C(z, z, z, 0) - C(z, z, z, z) = z(A - \beta(s_1 - s_0)^2(1 + \frac{z}{1-z} + \frac{z}{m+z} - \frac{z}{m} - 4z))$. Thus we have:

— if $1 - z - \frac{z}{2m} < \frac{m}{m+z} - \frac{z^2}{1-z}$, then $\beta(s_1 - s_0)^2(\frac{m}{m+z} - \frac{z^2}{1-z}) < A \leq \beta(s_1 - s_0)^2\frac{m}{m+z} \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (z, z, z, z)$;

— if $\frac{m}{m+z} - \frac{z^2}{1-z} < 1 - z - \frac{z}{2m} < \frac{m}{m+z}$, then $\beta(s_1 - s_0)^2(\frac{m}{m+z} - \frac{z^2}{1-z}) < A \leq \beta(s_1 - s_0)^2(1 - z - \frac{z}{2m}) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (0, 0, 0, 0)$, and $\beta(s_1 - s_0)^2(1 - z - \frac{z}{2m}) < A \leq \beta(s_1 - s_0)^2\frac{m}{m+z} \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (z, z, z, z)$;

— if $\frac{m}{m+z} < 1 - z - \frac{z}{2m}$, then $\beta(s_1 - s_0)^2(\frac{m}{m+z} - \frac{z^2}{1+z}) < A \leq \beta(s_1 - s_0)^2\frac{m}{m+z} \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (0, 0, 0, 0)$.

When $A > \beta(s_1 - s_0)^2\frac{m}{m+z}$, $C(z, z, 0, 0) - C(z, z, z, 0) = z(A - \beta(s_1 - s_0)^2(1 + \frac{z}{m+z} - \frac{z}{m} - \frac{z}{1-z}))$, $C(z, z, z, 0) - C(z, z, z, z) = z(A - \beta(s_1 - s_0)^2(1 + \frac{z}{1-z} + \frac{z}{m+z} - 4z - \frac{z}{m}))$, $C(z, z, 0, 0) - C(z, z, z, z) = 2z(A - \beta(s_1 - s_0)^2(1 + \frac{z}{m+z} - \frac{z}{m} - 2z))$. One can check that $1 + \frac{z}{1-z} + \frac{z}{m+z} - 4z - \frac{z}{m} < 1 + \frac{z}{m+z} - \frac{z}{m} - 2z < 1 + \frac{z}{m+z} - \frac{z}{m} - \frac{z}{1-z}$. We conclude that:

— if $\frac{m}{m+z} < 1 + \frac{z}{m+z} - \frac{z}{m} - 2z$ (which is equivalent to $z < \frac{m(1-2m)}{1+2m}$), then $\beta(s_1 - s_0)^2\frac{m}{m+z} < A \leq \beta(s_1 - s_0)^2(1 + \frac{z}{m+z} - \frac{z}{m} - 2z) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (z, z, 0, 0)$, and $A > \beta(s_1 - s_0)^2(1 + \frac{z}{m+z} - \frac{z}{m} - 2z) \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (z, z, z, z)$.

— if $1 + \frac{z}{m+z} - \frac{z}{m} - 2z < \frac{m}{m+z}$, then $A > \beta(s_1 - s_0)^2\frac{m}{m+z} \Rightarrow (a_1^*, a_2^*, b_1^*, b_2^*) = (z, z, z, z)$;

Up to now, we have identified the optimal strategies based on the value of A . Let's reorganize and summarize them below.

— If $z \leq \frac{2m-1}{4m+2}$, then

$$(a_1^*, a_2^*, b_1^*, b_2^*) = \begin{cases} (0, 0, 0, 0), & \text{if } A \leq \beta(s_1 - s_0)^2\frac{1}{1+2z}; \\ (0, 0, z, z), & \text{if } \beta(s_1 - s_0)^2\frac{1}{1+2z} < A \leq \beta(s_1 - s_0)^2\left(1 - \frac{z}{m} - \frac{4z^2}{1+2z}\right); \\ (z, z, z, z), & \text{if } A > \beta(s_1 - s_0)^2\left(1 - \frac{z}{m} - \frac{4z^2}{1+2z}\right). \end{cases}$$

— If $z > \frac{2m-1}{4m+2}$ and $z > \frac{m(1-2m)}{1+2m}$, then

$$(a_1^*, a_2^*, b_1^*, b_2^*) = \begin{cases} (0, 0, 0, 0), & \text{if } A \leq \beta(s_1 - s_0)^2(1 - z - \frac{z}{2m}); \\ (z, z, z, z), & \text{if } A > \beta(s_1 - s_0)^2(1 - z - \frac{z}{2m}). \end{cases}$$

— If $z > \frac{2m-1}{4m+2}$ and $z \leq \frac{m(1-2m)}{1+2m}$, then

$$(a_1^*, a_2^*, b_1^*, b_2^*) = \begin{cases} (0, 0, 0, 0), & \text{if } A \leq \beta(s_1 - s_0)^2\frac{m}{m+z}; \\ (z, z, 0, 0), & \text{if } \beta(s_1 - s_0)^2\frac{m}{m+z} < A \leq \beta(s_1 - s_0)^2(1 + \frac{z}{m+z} - \frac{z}{m} - 2z); \\ (z, z, z, z), & \text{if } A > \beta(s_1 - s_0)^2(1 + \frac{z}{m+z} - \frac{z}{m} - 2z). \end{cases}$$